

Misallocation and Manufacturing TFP in China and India

By Chang-Tai Hsieh and Peter J. Klenow
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Presented By:

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Motivation

- Maximizing Every Dollar, A Utopia for Economic Efficiency.
- Misallocation, The Hidden Barrier.

- ✧ Revealing the Enigma, Insights from an Analysis of Manufacturing in China, India, and the U.S.
- ✧ Roadmap to the Utopia, How to get Closer?

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Introduction

Main Question

- How does resource misallocation impact aggregate TFP in China and India compared to the United States?

Previous Works

- **Productivity Analysis:** McKinsey Global Institute (1998). Klenow and Rodríguez-Clare (2005). Restuccia and Rogerson (2008).
- Technology, Trade, and Growth Models: Howitt (2000). Melitz (2003).
- Economic Disparity and Local Economies: William W. Lewis (2004). Banerjee and Duflo (2005).
- Business Cycle Accounting Method: Chari, Kehoe, and McGrattan (2007).
- Firm-Level Productivity and Efficiency: Foster, Haltiwanger, and Syverson (2008).

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This Paper

- Using microdata to quantify misallocation gaps in labor and capital across industries in China and India, and measuring their impact on Total Factor Productivity (TFP).

Model

Variables

Distortions:

- (τ_y) Affect K , L similarly e.g. taxes & subsidies, regulations, transportation.
- (τ_k) Raise the marginal product of capital relative to labor e.g. Access to credit.

Total Factor Productivity:

- $TFPQ \equiv A$: Measures the efficiency of factor inputs in generating output.
- $TFPR$: Accounts for distortions and firm-specific factors affecting productivity.

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Introduction to the Model

- Cobb-Douglas production technology for the final good:

$$Y = \prod_{s=1}^S Y_s^{\theta_s}, \text{ where } \sum_{s=1}^S \theta_s = 1. \quad (1)$$

- CES production technology for the industry output:

$$Y = \left(\sum_{s=1}^S Y_s^{\sigma} \right)^{\frac{1}{\sigma}}. \quad (2)$$

- Production function for each differentiated product:

$$Y_s = A_s K_s^{\alpha} L_s^{1-\alpha}.$$

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Firm's Problem

- Firm's Problem is to maximize its profit:

$$\pi_{si} = (1 - \tau_{Ysi})P_{si}Y_{si} - wL_{si} - (1 + \tau_{K_{si}})RK_{si}. \quad (4)$$

in which:

- $(1 - \tau_{Ysi})P_{si}Y_{si}$: Revenue adjusted for output distortions,
- wL_{si} : Cost of Labor,
- $(1 + \tau_{K_{si}})RK_{si}$: Cost of capital adjusted for capital distortions.

Resource Allocation and Productivity

- Marginal revenue products of labor and capital across firms depends on distortions.

$$MRPL_j = \frac{1}{1-\tau_{Y_j}} \frac{(1-\alpha_j)\sigma-1}{\sigma} \frac{P_j Y_j}{L_j} = w, \quad (5)$$

$$MRPK_j = \frac{1}{1+\tau_{K_j}} \frac{\alpha_j\sigma-1}{\sigma} \frac{P_j Y_j}{K_j} = R, \quad (6)$$

$$TFPR_j \propto (MRPK_j)^{\alpha_j} (MRPL_j)^{1-\alpha_j} \propto \frac{(1+\tau_{K_j})^{\alpha_j}}{1-\tau_{Y_j}}. \quad (7)$$

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Aggregate TFP and Misallocation

- When physical productivity ($A \equiv TFPQ$) and revenue productivity ($TFPR$) are jointly lognormally distributed, there is a simple closed-form expression for aggregate TFP , represented by:

$$\log(TFP_s) = \frac{1}{\sigma - 1} \log \left(\sum_{i=1}^{M_s} A_{si}^{\sigma-1} \right) - \sigma^2 \text{var}(\log(TFPR_{si})). \quad (8)$$

- The first part calculates the average productivity of all the companies in the sector.
- The second part of is reflecting the distortion.

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Data

Data

■ India:

Annual Survey of Industries (ASI) from 1987–1988 through 1994–1995 fiscal years conducted by the Indian government's Central Statistical Organisation.

■ China:

Annual Surveys of Industrial Production from 1998 through 2005 conducted by the Chinese government's National Bureau of Statistics.

■ United States:

Data from the Census of Manufactures (CM) from 1977, 1982, 1987, 1992, and 1997 conducted by the U.S. Bureau of the Census.

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Calibrating

Solving and Calibrating the Model

■ Capital distortion:

$$1 + \tau_{Ksi} = \frac{\alpha_s}{1 - \alpha_s} \cdot \frac{wL_{si}}{RK_{si}}. \quad (9)$$

■ Output distortion:

$$1 - \tau_{Ysi} = \frac{\sigma}{\sigma - 1} \cdot \frac{wL_{si}}{(1 - \alpha_s)P_{si}Y_{si}}. \quad (10)$$

■ Productivity (TFPQ):

$$A_{si} = \kappa_s \cdot \frac{(P_{si}Y_{si})^{\frac{\sigma-1}{\sigma}}}{K_{si}^{\alpha_s} \cdot L_{si}^{1-\alpha_s}} \text{ in which: } \kappa_s = \frac{w^{1-\alpha_s}(P_{si}Y_{si})^{-\frac{\sigma-1}{\sigma}}}{P_{si}}$$

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Results

A Utopian Counter Factual

Table 1: TFP Gain from Equalizing TFPR within Industries

China	1998	2001	2005
%	115.1	95.8	86.6
India	1987	1991	1994
%	100.4	102.1	127.5
United States	1977	1987	1997
%	36.1	30.7	42.9

Less Medium, More Extreme!

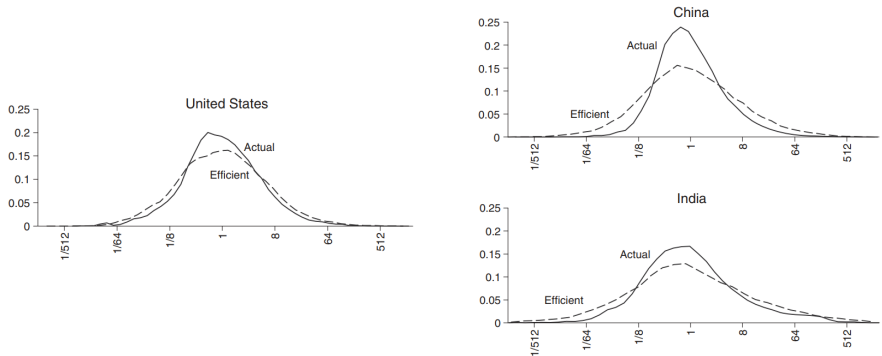


Figure 1: Distribution of Plant Size

As Productive as the Worst US!

Table 2: TFP Gains from Equalizing TFPR relative to 1997 U.S. gains

China	1998	2001	2005
%	50.5	37.0	30.5
India	1987	1991	1994
%	40.2	41.4	59.2

The Virtuous Cycle of TFP-Output

- Capital accumulation significantly amplifies the impact of TFP gains on manufacturing output.
- A 30% increase in TFP in China, could result in 69% increase in manufacturing output.
- A 59% increase in TFP in India, could result in 153% increase in manufacturing output.

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TFPR Distribution

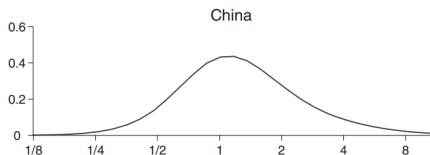
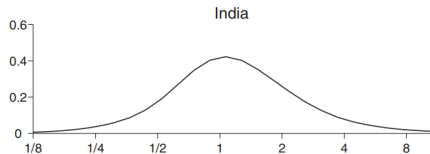
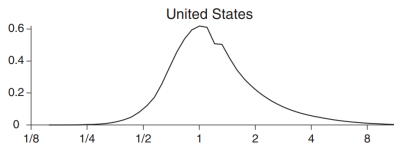


Figure 2: Distribution of TFPR

TFPQ Distribution

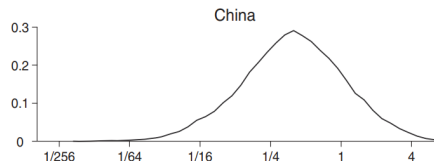
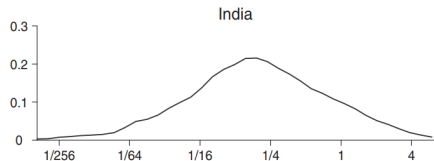
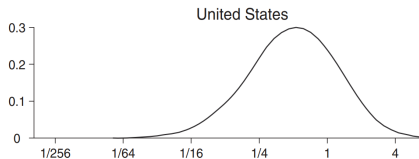


Figure 3: Distribution of TFPQ

Revealing the Enigma: TFPR in China

- The state-owned plants in China decreased from 29% in 1998 to 8% in 2005.
- SOE in China had 40% lower TFPR compared to private domestic plants.

Table 3: Regressions of Sector TFPR Dispersion on State Ownership in China

	1998	2001	2005	1998–2005
State ownership share	0.766 (0.165)	0.659 (0.153)	0.025 (0.213)	0.300 (0.080)
Year F.E.	NO	NO	NO	YES
Sector F.E.	NO	NO	NO	YES
N	406	403	407	3,237

Revealing the Enigma: TFPR in India

Table 4: Regression of Sector TFPR Dispersion on Delicensing and Size Restrictions in India

	(1)	(2)	(3)
Delicensed 1991	-0.298 (0.117)		-0.298 (0.117)
Delicensed 1991 $\times$ post-1991	0.032 (0.036)		-0.056 (0.040)
Size restriction		0.368 (0.173)	
Delicensed 1991 $\times$ post 1991 $\times$ size restriction			0.415 (0.120)

Conclusion

Conclusion

- The study examined misallocation of inputs in manufacturing plants in China (1998–2005) and India (1987–1994) compared to the United States (1977, 1987, 1997).
- Analysis revealed larger gaps in the marginal products of labor and capital across plants within industries in China and India compared to the United States.
- Simulations indicated that aligning with U.S. dispersion of marginal products could significantly increase TFP in both China (30%–50% increase) and India (40%–60% increase).

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Thank You!